SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I Vikram P 192511321(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: vikram

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.
- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5. A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.

- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

Question 1: Video Streaming Platform on the Cloud

Scenario: A media company needs to stream to millions of concurrent users with minimal buffering.

Cloud architecture (scalability and caching)

- **Ingest and transcoding:** Source video is uploaded to object storage (S3/Blob) and transcoded by a parallel pipeline (e.g., MediaConvert or FFmpeg on containers) into multiple bitrates for adaptive streaming (HLS/DASH).
- **Origin tier:** Object storage acts as the origin, fronted by an origin shield so that cache misses do not overload it.
- **Edge caching:** A global CDN (CloudFront/Akamai/Azure CDN) caches video segments close to viewers in multi-tier caches (edge, regional, origin shield). This offloads 90%+ of traffic from the origin.
- **Application tier:** Stateless microservices (login, catalogue, recommendations, playback) run in containers/VMs across multiple Availability Zones behind load balancers.
- **Data and in-memory caching:** Managed NoSQL for sessions and watch history, Redis/Memcached for metadata and hot data, relational DB for billing.
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Load balancing and auto-scaling for smooth streaming

- **Load balancing:** Geo-DNS/anycast routes users to the nearest healthy region; Layer-7 load balancers distribute requests with health checks and remove failing nodes automatically.
- **Auto-scaling:** Scaling groups add or remove instances based on metrics (CPU, requests per second, concurrent streams). Scheduled and predictive scaling pre-warm capacity before known peaks such as new releases or live events.
- **Adaptive bitrate streaming:** The player switches quality according to bandwidth, which minimizes buffering even when the network is congested.
- **Resilience:** Multi-AZ deployment, retries and graceful degradation (e.g., disable recommendations before disabling playback).

Cost optimization strategies

- Maximize CDN cache hit ratio to cut origin compute and egress costs; negotiate committed CDN pricing.
- Use reserved instances/savings plans for baseline load, on-demand for bursts, and spot instances for fault-tolerant transcoding.
- Tier storage (hot, infrequent access, archive) and delete unused renditions.
- Use efficient codecs (H.265/AV1) to reduce bandwidth, and right-size instances using monitoring data.

- Set budgets, tagging and alerts (FinOps) to track cost per stream-hour.

Question 2: Containers Across Multiple Cloud Providers (Fintech)

Scenario: A fintech startup struggles with consistency and deployment management across clouds.

Proposed solution: container orchestration + multi-cloud strategy

- **Kubernetes everywhere:** Use managed Kubernetes (EKS/AKS/GKE) or a unifying platform (Rancher, OpenShift, Anthos) so every cloud exposes the same API.
- **Immutable images:** Build once, store in a central container registry and deploy the same image to every cluster to guarantee consistency.
- **Infrastructure as Code and GitOps:** Terraform provisions clusters; Helm charts and GitOps tools (Argo CD/Flux) keep deployments declarative, version-controlled and auditable.
- **Policy and security:** Policy-as-code (OPA/Kyverno), common IAM federation, secrets management (Vault) and image scanning in CI/CD, which matters for financial compliance.
- **Unified observability:** Prometheus/Grafana, centralized logging and tracing (OpenTelemetry).

Service discovery and scaling

- **Service discovery:** Kubernetes Services and CoreDNS inside a cluster; a service mesh (Istio/Linkerd) or Consul for cross-cluster discovery, mTLS, retries and traffic shifting; global load balancer/DNS for routing users across clouds.
- **Scaling:** Horizontal Pod Autoscaler (CPU, memory or custom metrics), Cluster Autoscaler to add nodes, Vertical Pod Autoscaler for right-sizing, and KEDA for event-driven scaling (e.g., queue depth).

Benefits and risks of multi-cloud

Benefits	Risks
Avoids vendor lock-in and improves bargaining power	Higher operational complexity and tooling overhead
Higher resilience if one provider fails	Skill gaps; need expertise in several platforms
Meets data-residency and regulatory needs by region	Inconsistent IAM/security models enlarge attack surface
Choose best-of-breed services and optimal pricing	Inter-cloud data transfer (egress) costs and latency
Geographic proximity to customers	Harder data consistency and fragmented monitoring

Question 3: Backup and Disaster Recovery (E-learning Platform)

Scenario: Accidental deletion caused data loss in the cloud; a robust backup/DR strategy is

needed.

Solution design (backup, replication, versioning)

- **Versioning:** Enable object versioning, soft-delete/recycle-bin and MFA-delete on storage so accidentally deleted or overwritten files can be restored.
- **Cloud-native backup:** Automated, policy-based backups (AWS Backup/Azure Backup) of databases, volumes and file shares, with point-in-time recovery (PITR) for databases.
- **Replication:** Cross-region replication of object storage and database replicas in a second region.
- **Immutability and isolation:** Object Lock/WORM backup vaults in a separate account/subscription, so a mistake or attacker cannot delete backups.
- **Access control:** Least-privilege IAM, approval workflow for delete operations, audit logs and alerts on bulk deletions.

How RTO and RPO are achieved

- **RPO:** Continuous replication/PITR for the student records DB (RPO of seconds to minutes), hourly or daily snapshots for course content.
- **RTO:** Infrastructure as Code to rebuild quickly, warm standby for critical services, automated failover (DNS health checks) and tested runbooks.

DR strategy	Typical RPO	Typical RTO	Cost
Backup and restore	Hours	Hours	Lowest
Pilot light	Minutes	Tens of minutes to hours	Low
Warm standby	Seconds to minutes	Minutes	Medium
Active-active multi-site	Near zero	Near zero	Highest

RPO (maximum tolerable data loss) is met by how often data is protected; RTO (maximum tolerable downtime) is met by how quickly the environment can be brought back.

Steps to ensure business continuity

- Detect the failure or deletion through monitoring and alerts.
- Declare an incident and activate the DR plan and communication tree.
- Fail over traffic to the standby region (or restore from version/snapshot for deletions).
- Validate data integrity and application functionality.
- Communicate status to students and staff.
- Fail back after the primary is repaired; perform a post-incident review.
- Test regularly through DR drills and update runbooks.

Question 4: Hybrid Cloud for a Government Agency

Scenario: Sensitive data and compliance-bound applications must stay on-premises while

others use the cloud.

Hybrid architecture with secure communication

- **Placement:** Classify data. Sensitive and regulated workloads stay in the on-premises private cloud; non-sensitive and elastic workloads run in the public cloud.
- **Connectivity:** Dedicated private links (Direct Connect/ExpressRoute) with redundant IPsec VPN as backup; hub-and-spoke network topology.
- **Network security:** Firewalls, segmentation, private endpoints, API gateway as the only controlled path, IDS/IPS, and zero-trust principles (never trust, always verify).
- **Consistent platform:** Common tools (Kubernetes, Azure Arc/Anthos/Outposts) give a single management plane.

Identity management and encryption

- **Identity:** One central identity provider federated to both environments through SAML/OIDC, with Active Directory sync, SSO, mandatory MFA, RBAC/ABAC, privileged access management and just-in-time access.
- **Encryption in transit:** TLS 1.2+ and mutual TLS between services; IPsec on links.
- **Encryption at rest:** AES-256 for storage and databases.
- **Key management:** Keys held in on-prem HSM/KMS (BYOK/HYOK) under agency control, with rotation, separation of duties and key-usage auditing.

Challenges in hybrid cloud management

- Integration complexity and latency between environments.
- Keeping security policies and compliance consistent in both places.
- Unified visibility, monitoring and logging.
- Data movement, synchronization and egress costs.
- Skill gaps and heterogeneous tooling.
- Governance, shadow IT and audit evidence across two environments.

Question 5: High Availability and Fault Tolerance (SaaS Provider)

Scenario: The application must keep running even if an entire region fails.

Multi-region deployment strategy

- **Topology:** Active-active (or active-passive for lower cost) deployment in at least two regions, each spread across multiple Availability Zones.
- **Traffic management:** Global load balancing (Route 53 latency/health-based routing, Azure Traffic Manager/Front Door, anycast) directs users to the nearest healthy region.
- **Stateless application tier:** Session data kept in shared stores so any region can serve any user; identical stacks deployed using Infrastructure as Code.
- **Supporting services:** CDN, replicated object storage and managed message queues in each region.

Data replication and failover configuration

- **Database replication:** Use multi-region managed databases (Aurora Global Database, DynamoDB Global Tables, Cosmos DB, Spanner). Asynchronous replication gives low latency with small RPO; synchronous gives zero data loss at higher latency.
- **Conflict handling:** For multi-writer setups use last-writer-wins or application-level conflict resolution; otherwise a single write region with promotable replicas.
- **Failover:** Health checks with low DNS TTL (30-60 s) detect regional failure and shift traffic automatically; replica is promoted to primary; failback is planned and controlled.
- **Testing:** Chaos engineering and scheduled failover drills confirm RTO/RPO targets.

Monitoring and alerting for proactive detection

- **Metrics:** Track golden signals (latency, traffic, errors, saturation) plus replication lag, per region.
- **Logs and traces:** Centralized logging and distributed tracing to find root causes.
- **Synthetic monitoring:** Probes from multiple regions simulate user journeys and detect outages before customers do.
- **Alerting:** SLO-based alerts, anomaly detection and escalation (PagerDuty/Opsgenie) with runbooks; tune thresholds to avoid alert fatigue.

Set 2

Question 1: Big Data Analytics for a Retail Company

Scenario: Terabytes of daily transactions; near real-time analysis for recommendations and inventory.

Cloud-based big data architecture

- **Sources:** POS systems, e-commerce site, mobile app, inventory systems.
- **Ingestion:** Streaming via Kafka/Kinesis/Event Hubs, change-data-capture from databases, scheduled batch loads from stores.
- **Distributed storage:** Data lake on object storage (S3/ADLS/GCS) organized in raw, curated and consumption zones, in columnar formats (Parquet/Delta/Iceberg).
- **Distributed processing:** Apache Spark for batch, Spark Structured Streaming/Flink for streams.
- **Serving layer:** Cloud data warehouse (Redshift/BigQuery/Snowflake/Synapse) for BI, Redis/NoSQL for low-latency recommendation lookups, BI dashboards and ML services.

Integrating batch and stream processing

- **Lambda architecture:** Speed layer gives near real-time views (e.g., low-stock alerts, live recommendations); batch layer periodically recomputes accurate full-history results; serving layer merges both.
- **Kappa/unified approach:** A single engine (Spark/Flink with Delta Lake) processes both streams and replays, simplifying code.
- Batch jobs retrain recommendation models nightly; streaming applies them to live events.

Data lakes and data warehouses together

- Land everything in the lake first, transform (ELT) and load curated data into the warehouse; this combined pattern is a "lakehouse".

Aspect	Data lake	Data warehouse
Data	Raw, semi/unstructured, all history	Cleaned, modelled, structured
Schema	Schema-on-read	Schema-on-write
Users	Data scientists, engineers	Analysts, business users
Cost	Low-cost storage	Higher cost, faster SQL

Scalability and cost efficiency

- Separate storage from compute and autoscale clusters; use spot instances for batch jobs.
- Partition and compress data (Parquet) to reduce scan cost.
- Lifecycle policies to tier old data to cheaper storage; use serverless queries for ad-hoc work.

- Materialized views and caching for frequent queries; right-size and shut down idle clusters.

Question 2: Secure Cloud for a Healthcare Organization

Scenario: Patient records and imaging migrate to the cloud under strict privacy regulations.

Secure cloud architecture

- **Isolation:** Private network (VPC/VNet) with private subnets; databases and storage have no public endpoints; private links to services; WAF at the edge.
- **Remote access:** Doctors connect through SSO with MFA over VPN/zero-trust access from multiple locations.
- **Data services:** Imaging (DICOM) and EHR stored in encrypted object storage and managed health-data services; sign a Business Associate Agreement and respect data-residency requirements.
- **Segmentation and minimization:** Separate production, analytics and development; de-identify data for analytics.

Encryption, IAM and key management

- **Encryption:** TLS 1.2+ in transit; AES-256 at rest for databases, images and backups; field-level encryption for highly sensitive identifiers.
- **IAM:** SSO + MFA, role-based and attribute-based access (role: doctor/nurse; attributes: department, treating relationship), least privilege, break-glass emergency access, session timeouts, privileged access management.
- **Key management:** Central KMS/HSM with customer-managed keys, envelope encryption, automatic rotation, separation between key admins and data admins, and logging of every key use.

Auditing, logging and compliance monitoring

- Log every access to patient data (who, what, when, from where) with CloudTrail/Cloud Audit Logs/Azure Monitor, stored in immutable storage.
- Send logs to a SIEM for correlation and alerts on unusual access patterns.
- Automated compliance checks (AWS Config/Security Hub/Azure Policy) against HIPAA, GDPR and India's DPDP Act requirements.
- Periodic access reviews, audits and retention of evidence.

Preventing unauthorized access and breaches

- Zero-trust model, MFA everywhere and conditional access (device posture, location).
- Network segmentation, IDS/IPS, DLP and regular vulnerability scanning and patching.
- User behavior analytics to detect compromised accounts; immutable backups against ransomware.
- Staff security training, penetration tests and a rehearsed incident-response plan.

Question 3: Serverless Architecture for Ticket Booking

Scenario: Sudden traffic spikes when popular events are announced; reduce infrastructure management.

Serverless architecture

- **Front end:** Static website on object storage + CDN.
- **API layer:** API Gateway handles authentication, throttling and caching.
- **Compute:** Functions (Lambda/Azure Functions/Cloud Functions) for event search, seat hold, booking and payment.
- **Data:** Managed serverless database (DynamoDB/Aurora Serverless) plus Redis for temporary seat locks.
- **Asynchronous processing:** Queue/event bus (SQS/SNS/EventBridge) feeds functions that confirm payment, issue tickets and send email/SMS.
- **Spike protection:** A virtual waiting room and rate limits during on-sales.

Event-driven computing and scalability

- Functions run only in response to events (HTTP request, queue message, database stream) and scale automatically from zero to thousands of concurrent executions.
- Components are decoupled; a slow payment step does not block browsing.
- Pay-per-invocation pricing means no cost when idle; provisioned concurrency can remove cold-start delay for critical paths.

Role of managed databases, API gateways and message queues

- **Managed databases:** Auto-scaling, built-in backups, replication and patching without administration.
- **API gateways:** Single secure entry point with authentication, rate limiting, request validation and caching.
- **Message queues:** Buffer spikes (load levelling), enable retries and dead-letter queues, and ensure no booking request is lost. Idempotency keys prevent double booking.

Advantages and limitations of serverless for this workload

Advantages	Limitations
Automatic, rapid scaling for traffic spikes	Cold-start latency for infrequently used functions
No server management; faster development	Execution time, memory and concurrency limits
Pay only for actual usage	Database connection limits and contention on hot seats
Built-in high availability	Vendor lock-in; harder debugging and testing
Lower operations effort	Can cost more than containers under constant heavy load

Question 4: ETL/ELT Analytics Platform for a Multinational

Scenario: Data from sales, finance, manufacturing and support must be centralized with consistent quality.

Cloud-based ETL/ELT pipeline

- **Extract:** Connectors/APIs, CDC and file feeds from ERP, CRM, MES and ticketing systems (Fivetran/Airbyte/AWS Glue/Azure Data Factory).
- **Load:** Land data unchanged in a raw zone of the data lake.
- **Transform:** ELT using warehouse compute or Spark/dbt: staging, then cleansed, then curated data marts (star schema) in a cloud warehouse (Snowflake/BigQuery/Redshift).
- **ETL vs ELT:** Use ETL when masking or heavy cleaning is needed before loading; ELT otherwise to exploit scalable warehouse compute.
- **Consume:** BI tools, ML, APIs.

Data quality, metadata and governance

- **Data quality:** Automated checks for completeness, validity, uniqueness, consistency and timeliness (Great Expectations, Deequ, dbt tests); quarantine bad records; master data management for common customer/product keys.
- **Metadata:** Central data catalog (Glue Catalog/Purview/DataHub/Collibra) with business glossary, technical schemas and end-to-end lineage.
- **Governance:** Named data owners and stewards, classification, RBAC with column masking, audit logging and compliance with GDPR and regional laws.

Workflow orchestration and scheduling

- Tools such as Apache Airflow, Azure Data Factory, AWS Step Functions or Dagster define pipelines as DAGs with dependencies.
- They provide scheduling, retries, alerting, SLAs, backfills and monitoring, and make pipelines idempotent and repeatable.

Optimizing query performance and storage costs

- Columnar formats (Parquet), compression, partitioning and clustering to prune scans.
- Incremental loads, materialized views, aggregate tables and result caching.
- Separate compute for ETL vs BI workloads; auto-suspend idle warehouses.
- Tier or expire old data and remove duplicates.

Question 5: ML Platform for Predicting Delivery Delays

Scenario: Historical shipments, weather and live GPS feed continuous model training and real-time predictions.

Cloud-based ML and analytics architecture

- **Ingestion:** Live GPS and weather via Kafka/Kinesis; historical shipments from the warehouse/lake.

- **Stream processing:** Flink/Spark Streaming computes live features (current speed, route deviation, congestion).
- **Feature store:** Offline store for training and online store for low-latency inference (SageMaker Feature Store/Feast/Vertex).
- **Training:** Distributed training in SageMaker/Azure ML/Vertex AI; model registry for versioning.
- **Serving and feedback:** Real-time endpoint (API) and batch scoring; predictions and actual outcomes flow back to the lake for retraining.

Big data technologies for training and inference

- The data lake holds years of shipment data; Spark prepares large training sets and supports distributed training (MLlib) and parallel hyperparameter tuning, with GPU clusters for deep models.
- Inference runs in real time through autoscaled endpoints using online features, and in batch for daily route planning.

Cloud-native AI/ML services for deployment and monitoring

- Managed training, AutoML, model registry and one-click deployment to autoscaling endpoints.
- Safe rollout through canary/A-B deployment and shadow testing; MLOps CI/CD pipelines.
- Model monitoring for data drift, concept drift and accuracy decay, with alerts that trigger retraining; explainability tools for transparency.

Strategies for ingestion, feature engineering and model updates

- **Ingestion:** Partitioned topics, schema registry, back-pressure handling and micro-batching to manage volume.
- **Feature engineering:** Reusable features in a feature store, time-window aggregates (average speed over 15 min), geospatial and weather joins; same logic for training and serving to avoid train-serve skew.
- **Model updates:** Scheduled and drift-triggered retraining, champion-challenger evaluation, versioning and rollback.

Set 3

Question 1: Migrating Legacy Applications (Manufacturing Company)

Scenario: Move legacy enterprise apps and databases from on-premises to the cloud with minimal downtime.

Migration strategy

- **Assess:** Discover applications, map dependencies, baseline performance, estimate TCO and classify each application using the 6 Rs.
- **Plan:** Build a secure landing zone (network, IAM, logging), define migration waves and rollback plans.
- **Pilot:** Migrate a low-risk application first to prove tooling and process.
- **Migrate in waves:** Use replication tools (AWS Migration Hub/DMS, Azure Migrate) with continuous data sync.
- **Optimize:** Right-size, tune costs and decommission old hardware.

Applying the six migration approaches

Approach	Meaning	Example here
Rehost	Lift-and-shift VMs unchanged	Stable legacy file/print servers moved quickly
Replatform	Minor optimization without redesign	Move on-prem Oracle/SQL DB to a managed database service
Refactor	Re-architect for cloud-native	Core ERP modules re-written as microservices to scale
Repurchase	Replace with SaaS	Move HR, CRM or email to a SaaS product
Retire	Decommission unused apps	Redundant reporting tools
Retain	Keep on-prem for now	Shop-floor control systems needing very low latency or with compliance limits

Validating performance after migration

- Capture baseline metrics before migration and compare response time, throughput, latency, CPU/memory and error rate afterwards.
- Run load and stress tests, database query benchmarks and network latency tests.
- Use APM and synthetic monitoring, user acceptance testing and security validation; right-size based on results.

Reducing migration risk and downtime

- Continuous replication (CDC) so the final cutover window is minutes.
- Blue-green or parallel run, with a tested rollback plan.

- Migrate in small waves during off-peak hours; lower DNS TTL before cutover.
- Take full backups, apply security controls from day one, and train staff with clear runbooks.

Question 2: Smart City IoT Analytics

Scenario: Thousands of sensors (traffic, air quality, energy, safety) require real-time processing.

Cloud-based architecture for IoT collection and analytics

- **Devices and edge:** Sensors send data over MQTT/CoAP/LoRaWAN to edge gateways that filter and aggregate locally.
- **Ingestion:** IoT Hub/IoT Core handles device registry, authentication and secure connectivity.
- **Messaging:** Message broker (Kafka/MQTT broker/Kinesis) buffers streams.
- **Processing:** Stream processing (Flink/Spark Streaming/Azure Stream Analytics) for real-time rules and anomaly detection.
- **Storage:** Hot path: time-series DB for dashboards and alerts. Cold path: data lake for history and ML.
- **Presentation:** City operations dashboards, alert notifications and open APIs.

Integration of brokers, stream processing and storage

- Sensors publish to topics; the broker decouples producers and consumers.
- The stream processor consumes events, applies windowing and aggregation, and detects conditions such as pollution thresholds or accidents.
- Results go to the time-series DB and alerting service; raw data is archived in cloud storage for batch analytics and model training.

Handling high-volume, high-velocity data

- Partitioned topics and horizontal scaling of consumers.
- Edge aggregation and downsampling, compression and batching to reduce load.
- Time-series databases with retention tiers; back-pressure handling.

Scalability, reliability and security

- **Scalability:** Autoscaling managed services and elastic partitions.
- **Reliability:** Multi-AZ deployment, durable replicated messages, retries and dead-letter queues.
- **Security:** Unique device identity (X.509 certificates), TLS, secure boot and OTA updates, least-privilege access, network segmentation, and anonymization of sensitive data such as camera feeds.

Question 3: Business Intelligence Dashboards (Global E-commerce)

Scenario: Executives need interactive dashboards with real-time and historical analytics.

Cloud-based BI architecture

- **Sources:** Order system, CRM, web clickstream, marketing platforms.
- **Ingestion:** Streaming (Kafka/Kinesis) for real time and batch ELT for history into a data lake.
- **Warehouse:** Cloud data warehouse (Snowflake/BigQuery/Redshift/Synapse) with star schema; real-time ingestion via Snowpipe/streaming inserts, or a real-time OLAP store (Druid/Pinot).
- **Semantic layer:** Consistent business metrics (revenue, conversion) defined once.
- **Visualization:** Power BI/Tableau/QuickSight/Looker for executives.

Integration of visualization tools with the warehouse

- Native connectors or JDBC/ODBC with SSO authentication.
- Live/DirectQuery mode for real-time views; import/extract mode (e.g., SPICE) for fast historical dashboards.
- A shared semantic model and row-level security applied consistently.

Optimizing dashboard performance on large datasets

- Pre-aggregated tables and materialized views; incremental refresh.
- Partitioning and clustering, star-schema modelling and optimized SQL.
- Result caching and in-memory extracts; limit visuals and high-cardinality filters per page.
- Scale warehouse concurrency during peak executive usage.

Security measures for reports and sensitive data

- SSO with MFA and role-based access; row-level and column-level security by region or department.
- Data masking, encryption at rest and in transit, private networking.
- Restrict export/sharing, apply data classification, and keep audit logs of report access.

Question 4: Performance Monitoring for an Online Gaming Company

Scenario: Fluctuating workloads cause high latency and poor performance at peak hours.

Monitoring and optimization solution

- **Instrumentation:** Agents and OpenTelemetry on game servers, matchmaking, API and database tiers.
- **Metrics:** CloudWatch/Azure Monitor/Prometheus for CPU, memory, network, player count and latency.
- **Logs and traces:** Centralized logs (ELK/Loki/CloudWatch Logs) and distributed tracing (X-Ray/Jaeger).
- **User view:** Real-user monitoring and synthetic tests; dashboards in Grafana with alerting.
- **Action:** Autoscaling and game-server fleet management driven by these metrics.

How metrics, logs and tracing identify bottlenecks

- **Metrics:** show what is wrong, e.g., p95/p99 latency rises and CPU saturates at 8 pm.
- **Logs:** explain why, e.g., timeouts and connection pool errors.
- **Traces:** show where in the request path (login, matchmaking, session server, database) time is spent.
- Correlating all three pinpoints the cause, e.g., latency spike means traces show a slow DB call, and metrics show connection pool exhaustion.

Resource optimization and capacity planning

- Autoscaling with target-tracking and scheduled policies for predictable daily peaks, plus predictive scaling.
- Right-size instances, use read replicas, caching and sharding, and queue non-critical work.
- Analyse historical trends, forecast demand, run load tests and keep 20-30% headroom.

Improving availability while controlling cost

- Multi-AZ/region deployment, health checks and self-healing; graceful degradation.
- Reserved capacity for baseline, on-demand for peaks, spot instances for stateless/batch tasks.
- Scale down off-peak, use CDN for assets, tag resources and set budget alerts (FinOps).

Question 5: Data Lifecycle Management (Financial Institution)

Scenario: Petabytes of transaction data must be retained for compliance while keeping storage costs low.

Cloud-based data lifecycle strategy

Tier	Typical age (example)	Storage class	Access speed
Hot	0-90 days	Standard/SSD	Milliseconds
Warm	90 days - 1 year	Infrequent Access	Milliseconds
Cold	1 - 7 years	Archive (e.g., Glacier)	Minutes to hours
Deep archive	7+ years / legal hold	Deep archive	Hours

Lifecycle: create/ingest, active use, infrequent use, archive, retention expiry, secure deletion. Automated lifecycle policies move data between tiers.

Classification, retention and archival

- **Classification:** Label data by sensitivity (public, internal, confidential, restricted PII/financial) using tags and automated discovery (Macie/Purview).
- **Retention policies:** Retention periods set per regulation (commonly several years for financial records; confirm exact periods with compliance), with legal-hold support.

- **Archival:** Compressed columnar formats, WORM/Object Lock for immutability, a metadata catalog indexing archived data for retrieval, and certified secure deletion at expiry.

Governance, auditing and compliance

- Defined data owners and stewards, policies, lineage and a catalog.
- Audit logs of all access and changes, retained as evidence.
- Compliance with regulations such as PCI DSS, SOX, GDPR and central-bank/regulator guidelines avoids penalties and protects reputation; address data residency and privacy rights.

Balancing accessibility, security and cost

- Intelligent tiering based on access patterns; compression and de-duplication.
- Encryption (KMS) and RBAC at every tier; replication for durability.
- Define retrieval SLAs (expedited vs bulk) and use query-in-place on archives (e.g., Athena).
- Track storage cost and retrieval fees with budgets and reports.